

Chemistry

Group A:

Types of Properties & Solution:

- Colligate Properties (Def), what are the properties.
- Raoult's Law (with derivation).
- Relation between boiling.
- Herry's law with expression + limitation.
- Van Hoff Isotherm.
- Different between Ideal and real solutions.
- How to Calculate the molecular mass of solve.

Electro Chemistry :

- Farade's Law.
- Conductance, (Specific, equivalent) -def.
- Math
- Arhenius Theory of ionization.
- Hittorf's law. – Rule & Math.
- Transport Number. Def + derivation.
- Electrolytes – def
- Diff between Ionization vs dissociation.

Group B:**Chemical Equilibrium:**

- Forward Reaction, Backward Reaction, Chemical Equilibrium. Deffi.
- Explain Chemical Equilibrium is a dynamic Process.
- Condition for Chemical equilibrium.
- Characteristics of Chemical equilibrium.
- Law of Mass Action.
- Relation between K_p K_c .
- Le-Chatelier Principle.
- Math 2, 1

Chemical Kinetics:

- Zero , Pseudo Order Deffi & Example.
- Difference between Order & Molecularity.
- Integrated Equation for the 1st order and 2nd order Reaction rate with Expression.

Surface Chemistry & Colloids:

- Colloid - def, types. Colloids systems example.
- Colloids and adsorption.
- Lyophilic and lyophobic system Compare.
- Sols - def
- Comparison of physical adsorption
- Emulsions, type, example & rules.
- Isotherm with plots.
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